

Past year
Test 2 Chemistry II
2019/2020

Question 1

1		H_2CO_3 and HCO_3^-
		CH_3COOH and CH_3COO^-
		NH_4^+ and NH_3
		HPO_4^{2-} and H_2PO_4^-
2	i	$\text{C}_5\text{H}_7\text{O}_4\text{COOH (aq)} + \text{H}_2\text{O (l)} \rightleftharpoons \text{C}_5\text{H}_7\text{O}_4\text{COO}^- \text{ (aq)} + \text{H}_3\text{O}^+$ (aq) or $\text{C}_5\text{H}_7\text{O}_4\text{COOH (aq)} \rightleftharpoons \text{C}_5\text{H}_7\text{O}_4\text{COO}^- \text{ (aq)} + \text{H}^+ \text{ (aq)}$
	ii	A weak acid which changes color over a specific pH range in solution

Question 1

3	i	$\text{pOH} = 14 - \text{pH} = 14 - 11.33 = 2.67$ $[\text{OH}^-] = 10^{-2.67} = 2.13 \times 10^{-3} \text{ mol/L}$ $[\text{OH}^-] = x = [\text{C}_{10}\text{H}_{15}\text{ONH}^+] = 2.13 \times 10^{-3} \text{ mol/L}$ $[\text{C}_{10}\text{H}_{15}\text{ON}] = 0.035 - 2.13 \times 10^{-3} = 0.03287 \text{ mol/L}$
	ii	$K_b = (2.13 \times 10^{-3})^2 / 0.03287$ $= 1.38 \times 10^{-4}$
	iii	$K_a = K_w / K_b$ $= (1 \times 10^{-14}) / (1.38 \times 10^{-4})$ $= 7.25 \times 10^{-11}$
4		$0.67 = (x / 0.8) \times 100$ $x = [\text{H}^+] = 5.36 \times 10^{-3}$ $\text{pH} = -\log 5.36 \times 10^{-3} = 2.27$
5		One mole of sulphuric acid produces 2 moles of H^+ $[\text{H}^+] = 0.005 \times 2 = 0.01 \text{ M}$
		$\text{pH} = -\log [0.01] = 2$

Question 1

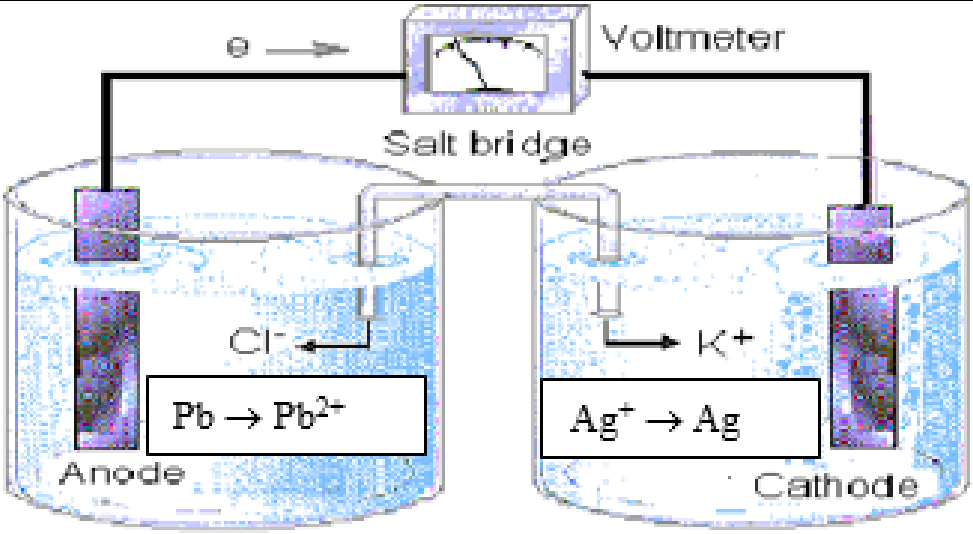
6	a	At equivalence point number of mol of H_3O^+ = number of mol of OH^- $n_{\text{HCl}} = n_{\text{NaOH}}$ $(MV)_{\text{HCl}} = (MV)_{\text{NaOH}}$ $(25\text{mL})(0.1\text{M}) = (0.1\text{M})V_{\text{NaOH}}$ $V_{\text{NaOH}} = 25\text{mL}/0.025\text{L}$																
	b(i)	<table><tr><td></td><td>$\text{HCl} + \text{H}_2\text{O}$</td><td>$\rightarrow \text{H}_3\text{O}^+(\text{aq})$</td><td>$\text{Cl}^-(\text{aq})$</td></tr><tr><td>Before</td><td>0.1 M</td><td>0</td><td>0</td></tr><tr><td>After</td><td>-0.1M</td><td>+0.1</td><td>+0.1</td></tr><tr><td>Change</td><td>0</td><td>0.1</td><td>0.1</td></tr></table> $[\text{H}_3\text{O}^+] = 0.1\text{M}$ $\text{pH} = -\log(0.1\text{M})$ $\text{pH} = 1$		$\text{HCl} + \text{H}_2\text{O}$	$\rightarrow \text{H}_3\text{O}^+(\text{aq})$	$\text{Cl}^-(\text{aq})$	Before	0.1 M	0	0	After	-0.1M	+0.1	+0.1	Change	0	0.1	0.1
	$\text{HCl} + \text{H}_2\text{O}$	$\rightarrow \text{H}_3\text{O}^+(\text{aq})$	$\text{Cl}^-(\text{aq})$															
Before	0.1 M	0	0															
After	-0.1M	+0.1	+0.1															
Change	0	0.1	0.1															
	b(ii)	<table><tr><td></td><td>HCl</td><td>NaOH</td><td>$\rightarrow \text{NaCl}(\text{aq})$</td></tr><tr><td>Before</td><td>0.0025 M</td><td>0.00125</td><td>0.00125</td></tr><tr><td>After</td><td>-0.00125M</td><td>-0.00125</td><td>+0.00125</td></tr><tr><td>Change</td><td>0.00125</td><td>0</td><td>0.00125</td></tr></table> $[\text{H}_3\text{O}^+] = 0.00125\text{mole}/0.0375\text{L} = 0.0333\text{M}$ $\text{pH} = 1.47$		HCl	NaOH	$\rightarrow \text{NaCl}(\text{aq})$	Before	0.0025 M	0.00125	0.00125	After	-0.00125M	-0.00125	+0.00125	Change	0.00125	0	0.00125
	HCl	NaOH	$\rightarrow \text{NaCl}(\text{aq})$															
Before	0.0025 M	0.00125	0.00125															
After	-0.00125M	-0.00125	+0.00125															
Change	0.00125	0	0.00125															
	b(iii)	Since the volumes and concentrations of the acid and base solutions are the same: $n(\text{H}^+) = n(\text{OH}^-)$, and $\text{pH} = 7.0$																

Question 2

1	i	B = +3
	ii	Co = +2
	iii	W = +3
2		Oxidation : $2\text{Ag} \rightarrow \text{Ag}_2\text{O}$ Reduction : $\text{Zn}^{2+} \rightarrow \text{Zn}$ Balancing O : Oxidation : $2\text{Ag} + \text{H}_2\text{O} \rightarrow \text{Ag}_2\text{O}$ Balancing H : Oxidation : $2\text{Ag} + \text{H}_2\text{O} \rightarrow \text{Ag}_2\text{O} + 2\text{H}^+$ Balancing OH^- : Oxidation : $2\text{Ag} + \text{H}_2\text{O} + 2\text{OH}^- \rightarrow \text{Ag}_2\text{O} + 2\text{H}^+ + 2\text{OH}^-$ Balancing e^- : Oxidation : $2\text{Ag} + 2\text{OH}^- \rightarrow \text{Ag}_2\text{O} + \text{H}_2\text{O} + 2e^-$ Reduction : $\text{Zn}^{2+} + 2e^- \rightarrow \text{Zn}$ Summation / Overall equation : $2\text{Ag}_{(s)} + 2\text{OH}^-_{(aq)} + \text{Zn}^{2+}_{(aq)} \rightarrow \text{Ag}_2\text{O}_{(aq)} + \text{H}_2\text{O}_{(l)} + \text{Zn}_{(s)}$

Question 2

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3	i	 Correct labelling: Cathode Anode Saltbridge Voltmeter Electron flow
	ii	$E^0_{\text{cell}} = E^0_{\text{red cathode}} - E^0_{\text{red anode}}$ $= 0.80 - (-0.13)$ $= 0.93 \text{ V}$
	iii	$\text{Pb(s)} \text{Pb}^{2+}(\text{aq}) \text{Ag}^{+}(\text{aq}) \text{Ag(s)}$
4		Pb^{2+}

682

614

206

738 X

~~870~~

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